

Empirical Methods for Law and Economics

European Doctorate in Law and Economics (EDLE), University of Bologna

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Lecture Notes: Teams or [Teaching Material](#)

1 Overview

This introductory doctoral course provides the statistical and econometric tools needed to read, evaluate and carry out empirical and experimental research on legal questions. No previous statistics or econometrics is assumed; some mathematical knowledge is required. Every meeting combines concepts with practical work in Stata, and every application comes from Law and Economics: contract enforcement, sanctions, prosecutorial independence and legal institutions.

Format. Eight teaching meetings of 3 hours each (including at least one break), followed by two 3-hour meetings for presentations and assessment: 30 hours in total.

2 Learning outcomes

By the end of the course, students will be able to:

1. describe variables and distributions, stating units, denominators and the unit of observation;
2. interpret sampling uncertainty, standard errors, confidence intervals and basic statistical tests;
3. estimate and interpret simple and multiple OLS regressions, including indicator (dummy) variables;
4. discuss regression assumptions and distinguish association from causation;
5. explain random assignment and the basic design of experiments;
6. keep a reproducible analysis in a Stata do-file;
7. write an accurate and constructive referee report on an empirical paper.

3 Programme

Meeting	Concepts	Practice and paper discussion
1	Research questions, unit of observation, measurement, variable types, descriptive statistics, missing values	Stata: open, inspect, tabulate and summarize World Bank data on contract enforcement; histograms
2	Sampling, standard deviation vs standard error, sampling distributions, confidence intervals	Stata: supplied simulation. Paper: Gneezy and Rustichini (2000), first reading
3	Relationships between variables, correlation, hypothesis tests for means and proportions	Stata: comparisons and tests; association vs causation
4	Simple OLS: coefficients, predictions, residuals, indicator variables	Stata: fit and interpret models. Paper: Voigt and Wulf (2019), first reading
5	Multiple OLS: controls, indicators, model building	Stata: compare models; conditional associations
6	OLS assumptions, uncertainty, assessing regression output	Stata: model assessment, changing samples. Paper: Voigt and Wulf (2019), second reading
7	Experimental economics: random assignment, incentives, internal and external validity	Classroom vignette and simulated data; evaluating an experimental design
8	Interpreting empirical results; writing referee reports	Stata: model comparison. Paper: Gneezy and Rustichini (2000), second reading; drafting comments
9–10	Individual presentations and discussion of the agreed papers	

4 Required readings

Paper discussions take place in meetings 2, 4, 6 and 8. Read only the indicated parts (30–45 minutes each) and bring short notes on the questions in the reading schedule distributed with the materials of each week. Reading notes are not graded.

Meeting	Paper	What to read
2	Gneezy, U. and Rustichini, A. (2000). A fine is a price. <i>Journal of Legal Studies</i> 29(1): 1–17. doi:10.1086/468061. Free PDF	Abstract; Sections I–II; Table 1 (p. 6) and Figure 1 (p. 7). Skip the formal tests and the model.
4	Voigt, S. and Wulf, A. J. (2019). What makes prosecutors independent? Analysing the institutional determinants of prosecutorial independence. <i>Journal of Institutional Economics</i> 15(1): 99–120 (first online 2017). doi:10.1017/S1744137417000212	Abstract; Sections 1 and 5; Table 1; column (1) of Table 2.
6	Voigt and Wulf (2019), second reading	Sections 7 and 8; all columns of Table 2 and footnotes.
8	Gneezy and Rustichini (2000), second reading	Section II, Table 1 and Figure 1 again; Sections IV.A, IV.D and V; Appendix A.

5 Assessment

Assessment is individual: a written referee report (70%) and an oral presentation with questions (30%).

Choice of the work. Each student chooses a working paper, or a seminar or workshop presentation, in agreement with the instructor. It must contain at least one empirical result that can be discussed with the tools of this course.

Written report (70%). 1,500–2,000 words in English, excluding references. It must contain:

1. an accurate summary: research question, contribution, data, unit of observation, sample, method and main findings;
2. the explanation of one central empirical result, citing the exact table, figure or slide, with magnitude, units, comparison, sample size and uncertainty where reported;
3. two or three specific substantive comments, each stating the issue, why it matters for a conclusion, and a feasible clarification or check;
4. an overall assessment with at least one strength and the main limitation.

For a seminar, distinguish information that was not available to you from evidence that the study lacks a feature. A replication and advanced econometric methods are not required.

Oral presentation and discussion (30%). In meetings 9–10, each student gives a presentation of about 20 minutes, followed by about 15 minutes of questions from the class. Present the question, one central result, a strength and your most important comment. Be ready to explain the statistical concepts involved and one Stata output from the course practice. The 30% is split into two components:

- *Presentation and answers (20%)*: clarity, correct use of the statistical concepts, and quality of the answers to questions.
- *Questions to colleagues (10%)*: in each presentation other than your own, you are expected to ask or contribute at least one question. The instructor records them and grades only your best three. A good question is accurate, specific to the paper's evidence (a table, a sample, an identification step) and constructive. The number of questions does not matter beyond the three that count; generic or repetitive questions earn no points.

Milestones (formative, not graded):

- meeting 3: the chosen paper or seminar is agreed with the instructor;
- meeting 5: one-page outline (question, data, method);
- meeting 7: draft comments;

- meeting 8: feedback and revision in class;
- before meeting 9: submission of the final report, by a common deadline communicated by the instructor.

Cite all sources consulted and disclose any assistance, including AI tools, according to the University's rules. Every student must be able to explain their report orally.

6 Practical information

- Bring a laptop with Stata installed (version 16 or later; University of Bologna licence or personal licence).
- Before each week, in Stata, set the working directory to the unpacked folder (`cd "path/to/folder"`) and keep the subfolders `data/` and `do/` together.
- Student do-files are worksheets completed in class; supplied demonstration scripts require no programming knowledge. Simulated data are always labelled as such.
- Slides, worksheets and reading questions are made available before each weekly block.

7 Reference books

No textbook is required; specific sections are indicated week by week.

- Wooldridge, J. M., *Introductory Econometrics: A Modern Approach*, Cengage (6th or 7th edition).
- Angrist, J. D. and Pischke, J.-S., *Mastering 'Metrics: The Path from Cause to Effect*, Princeton University Press (2015).
- Studenmund, A. H., *Using Econometrics: A Practical Guide*, Pearson (7th edition, 2017).
- Acock, A. C., *A Gentle Introduction to Stata*, Stata Press (any recent edition).
- Optional, free: Diez, D., Çetinkaya-Rundel, M. and Barr, C. D., *OpenIntro Statistics*, 4th edition, <https://www.openintro.org/book/os/>. Useful background for meetings 1–3.